

Smart IoT-Enabled Automated WeighBatching System

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Name of Industry:
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2. AIM and OBJECTIVES

Aim:

- To design and implement a Smart IoT-Enabled Automated Weigh Batching System for plastering applications using Teensy 4.1 and ESP32 for real-time monitoring and automated material control.

Objectives:

- To automate Weigh batching operations.
- To improve batching accuracy and consistency.
- To reduce manual errors and material wastage.
- To measure cement, sand, and water quantities accurately.
- To implement real-time monitoring using IoT technology.
- To control valves automatically using microcontroller-based logic.
- To develop a low-cost and efficient industrial batching solution.
- To improve plaster quality using proper mix ratio control.

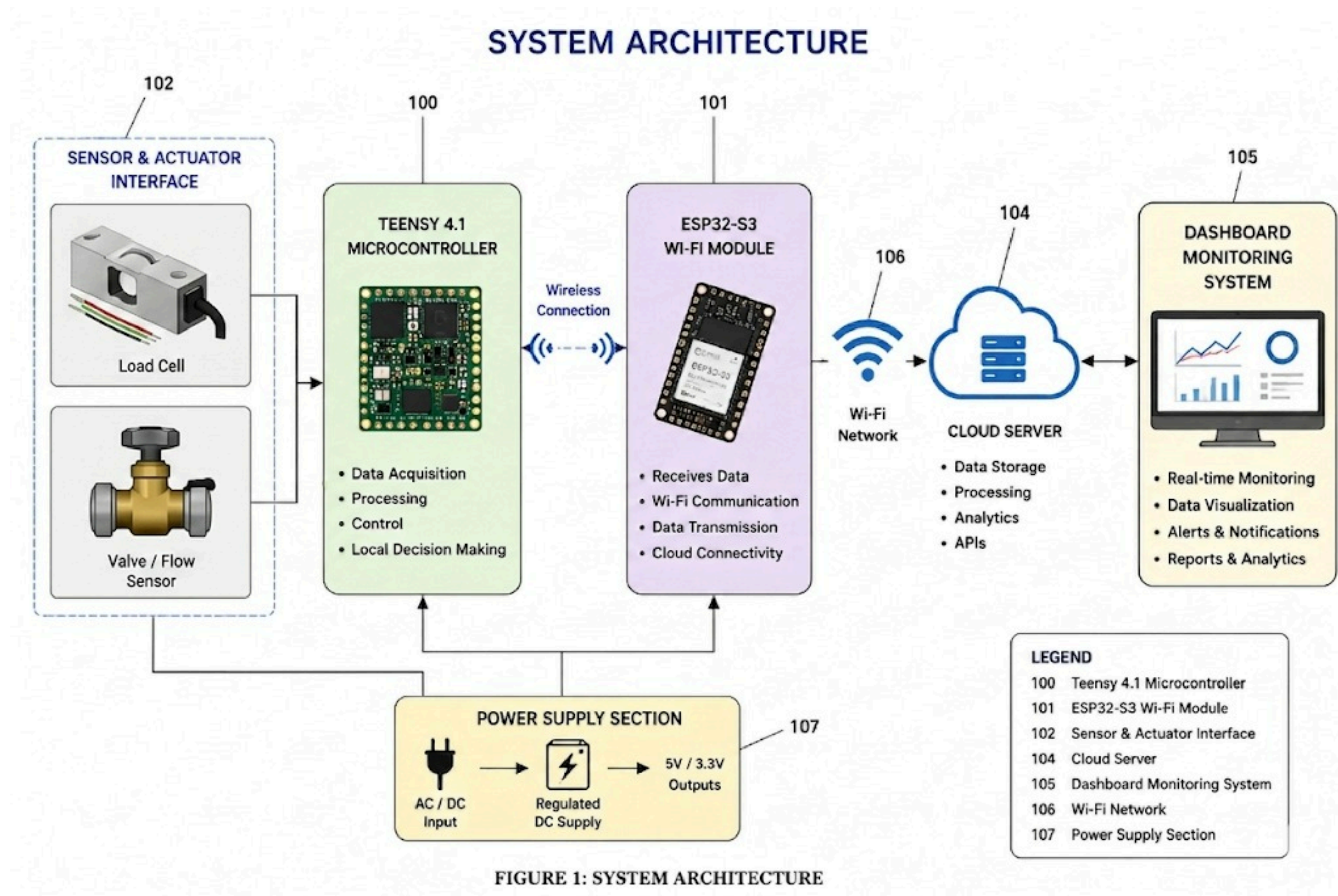
3. HARDWARE MATERIALS ARE REQUIRED AND SPECIFICATIONS

Sr. No	Component	Specification	Purpose
1	Teensy 4.1 MCU	ARM Cortex-M7	Main controller
2	ESP32 Module	Wi-Fi + BLE	Monitoring communication
3	ADS1232 ADC	24-bit Precision ADC	Load cell interfacing
4	Load Cells	4 × 500 kg	Weight measurement
5	Water Flow Sensor	YF-S201	Water monitoring
6	Solenoid Valves	12V DC	Water flow control
7	Butterfly Valve	Pneumatic Type	material mixture discharge
8	Power Supply	5V / 12V DC	System power
9	PCB/Perfboard	General Purpose	Hardware assembly

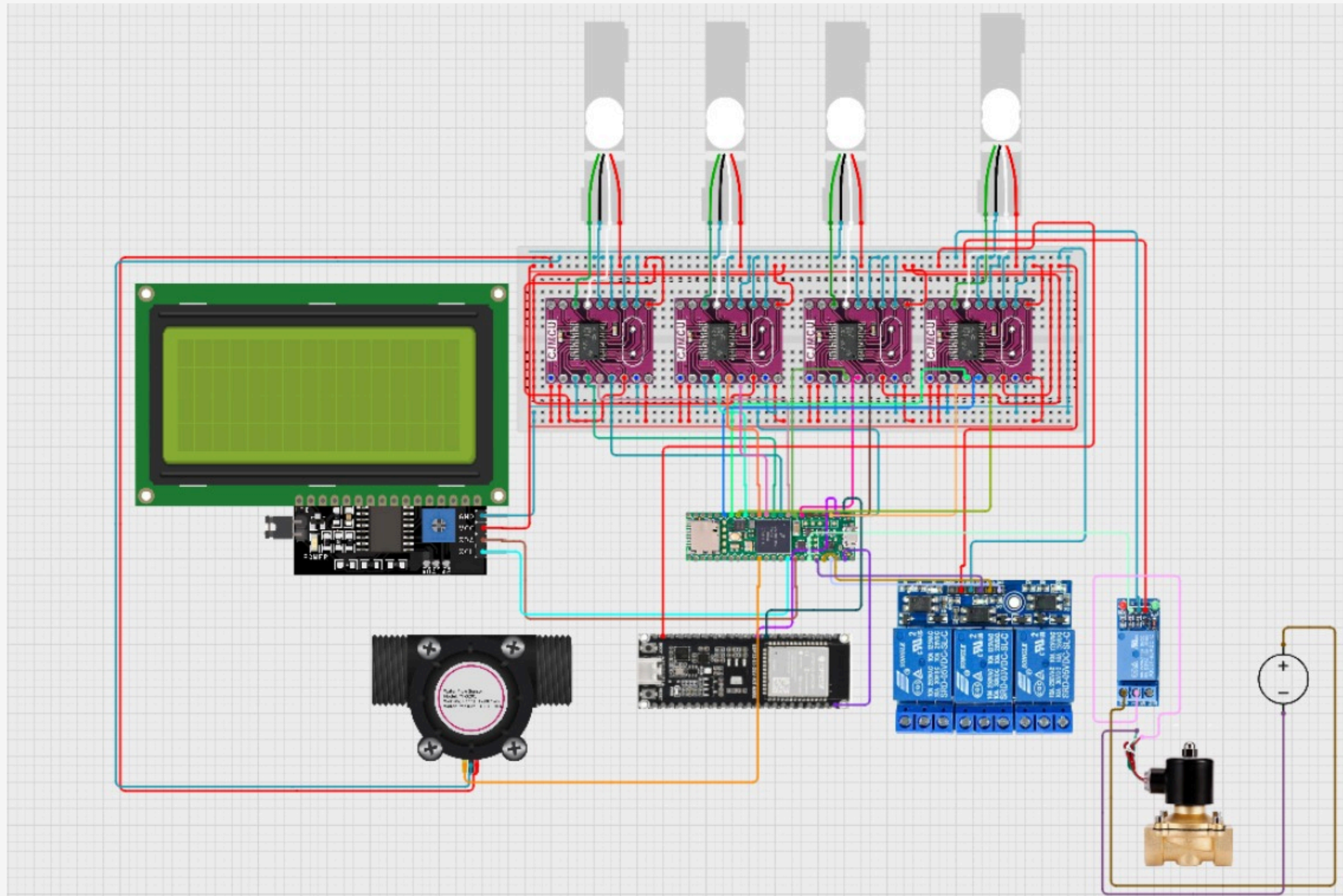
Specifications:

- Total Load Capacity: 2000 kg
- Communication: UART Serial Communication
- Wireless Monitoring: Wi-Fi
- ADC Resolution: 24-bit
- Operating Voltage: 5V / 12V DC

4. DESIGN (BLOCK DIAGRAM)



5. INTERFACING DIAGRAM

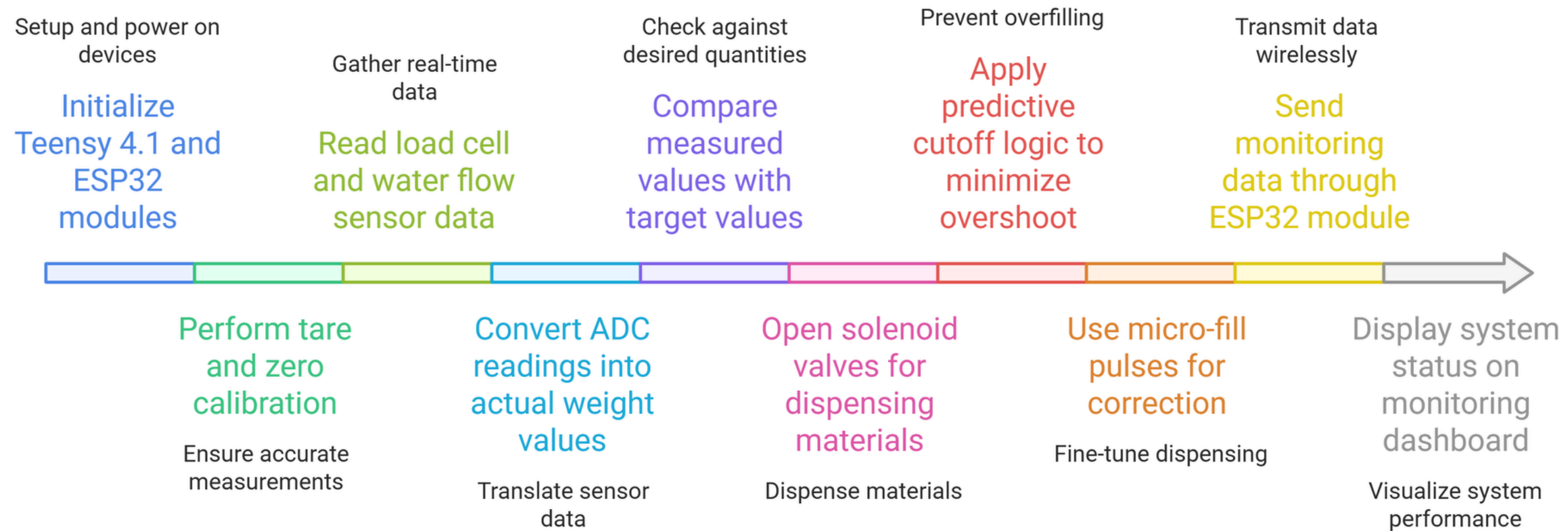


6. WORKING DESCRIPTION

- The system starts by initializing the controller, sensors, valves, and communication modules. Tare and zero calibration are performed before each batching cycle to remove offset errors.
- The load cells continuously measure the weight of cement and sand while the flow sensor monitors water quantity. The Teensy 4.1 processes the sensor data and compares it with predefined target values.
- Based on the calculated values, the controller automatically operates the solenoid valves for accurate material dispensing. Predictive cutoff logic is used to reduce material overshoot, and micro-fill pulse control improves batching precision.
- The ESP32 module transmits real-time batching information to the monitoring dashboard through Wi-Fi. The monitoring system displays batch quantity, material usage, and system status continuously.
- The automated system improves batching speed, consistency, and plaster quality while reducing manual operation and material wastage.

7. ALGORITHM and PROGRAM IMPLEMENTATION

Automated Batching System Operation

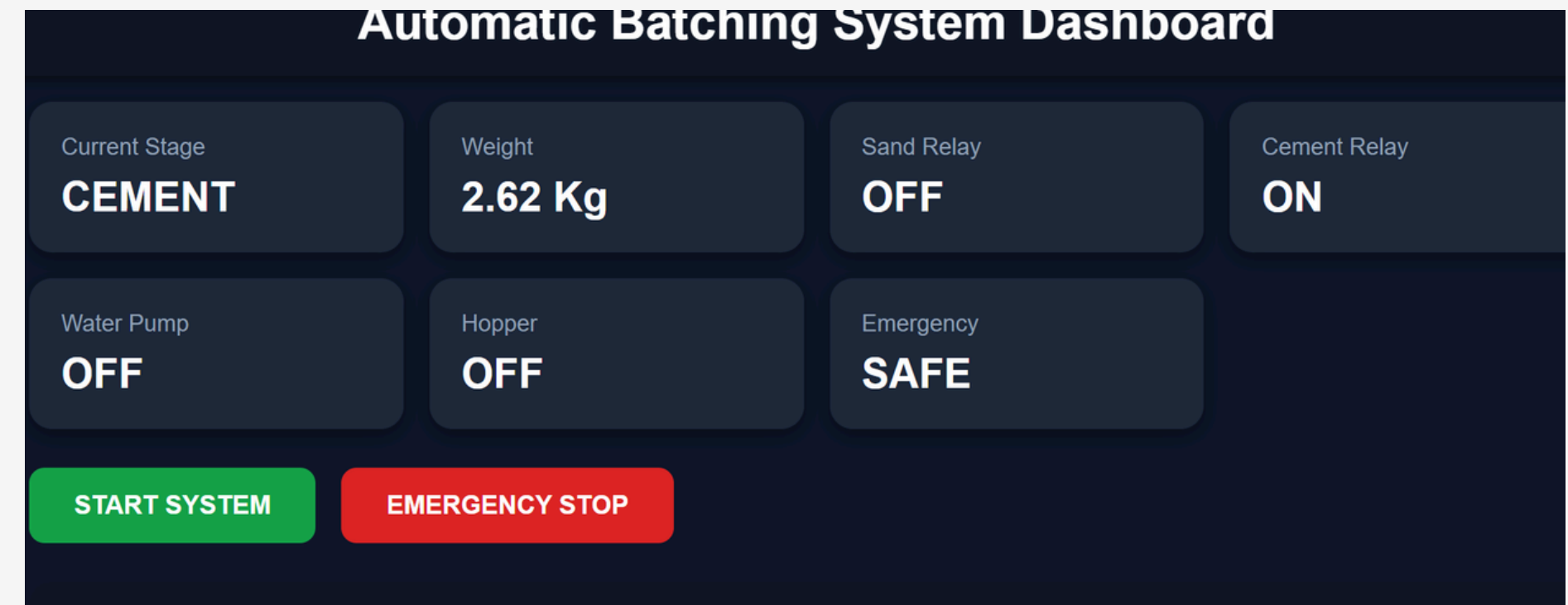
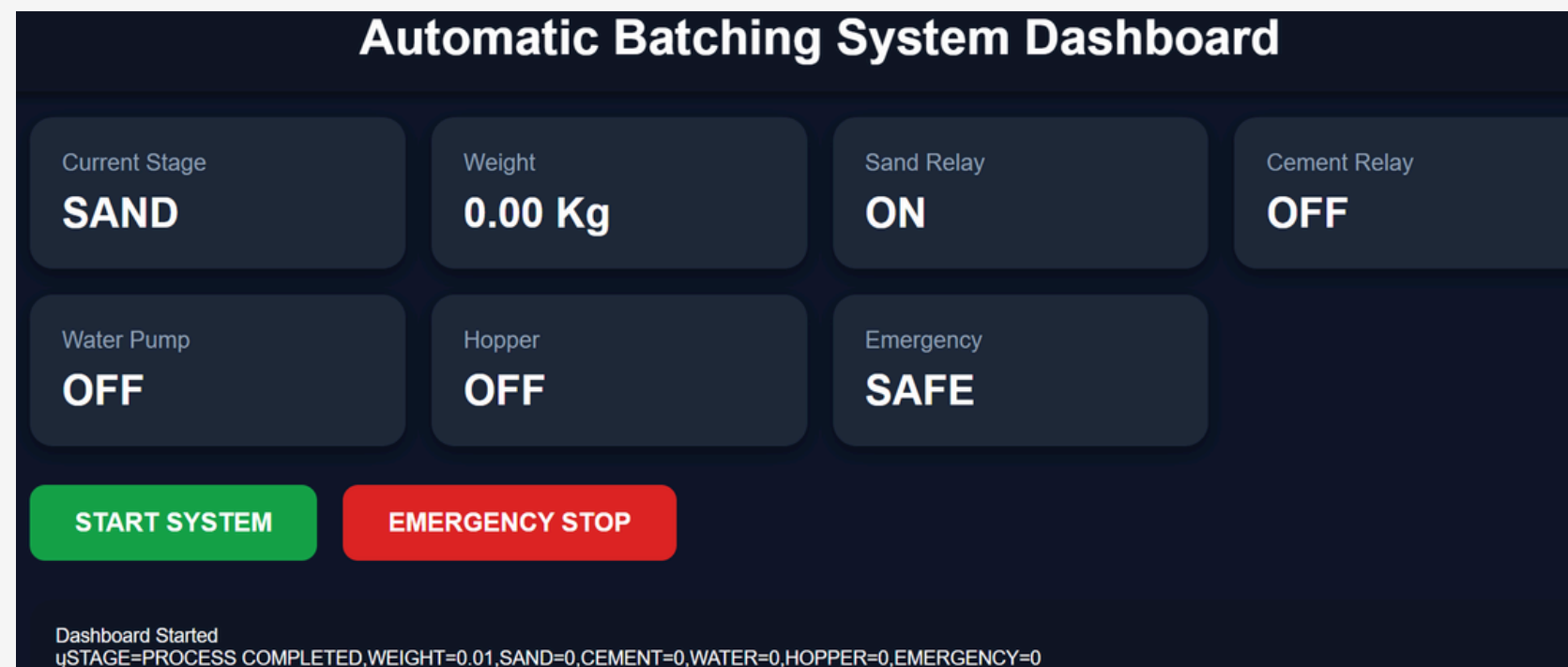
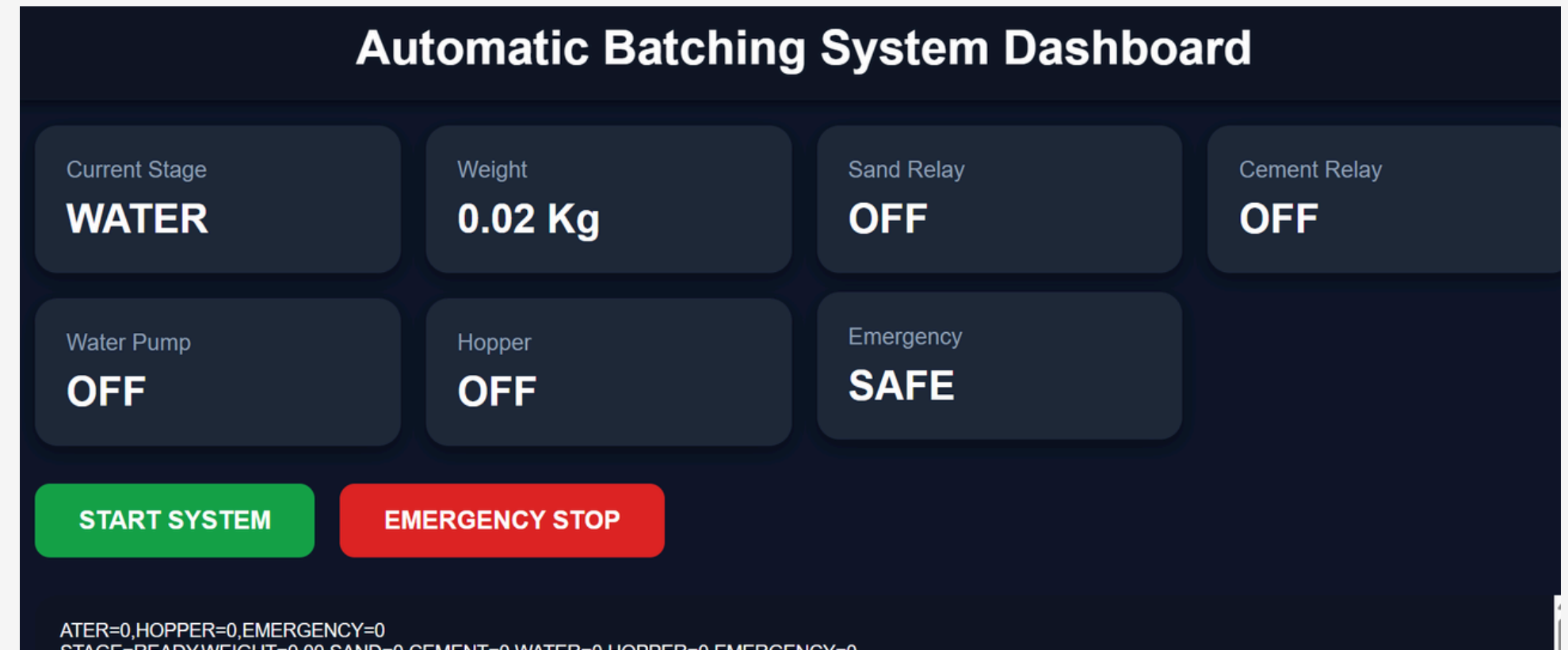
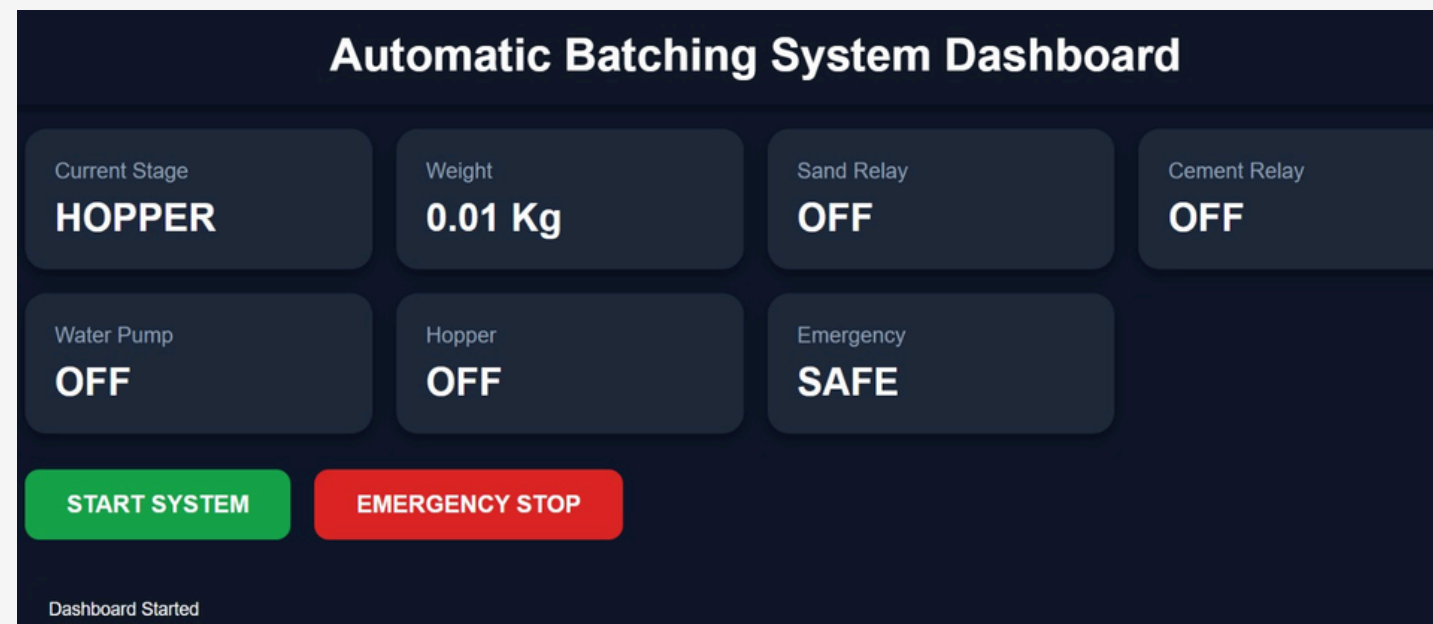


Program Implementation

- Embedded C/C++ programming used in Arduino IDE.
- UART communication implemented between Teensy and ESP32.
- Moving-average digital filtering used for noise reduction.
- Automated valve control logic implemented successfully.
- Real-time monitoring communication achieved through Wi-Fi.

9. RESULTS AND CONCLUSION

Results:



9. RESULTS AND CONCLUSION

Conclusion:

- The Smart IoT-Enabled Automated Weigh Batching System provides accurate, efficient, and reliable mortar batching for plastering applications. The integration of real-time sensing, automated control, and wireless monitoring improves batching consistency and reduces manual errors.
- The system acts as a compact and cost-effective alternative to conventional industrial batching systems and can be further expanded for future smart construction applications.

10. REFERENCES

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